
WEEK6:-TASK4

MILKA NAVEENA

TASK 4 — MODEL DEVELOPMENT & TRAINING

4.1 Define X and y

```
X = df[features]
y = df["performance"]
```

Encode:

```
X = pd.get_dummies(X, drop_first=True)

label_encoder = LabelEncoder()
y = label_encoder.fit_transform(y)
```

4.2 Split Dataset

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

Check:

```
print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])
```

Output

```
Training samples: 316
Testing samples: 79
```

4.3 Select Machine Learning Algorithm

Random Forest Classifier

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)
```

4.4 Why Random Forest?

I selected Random Forest because:

- It works well with tabular data.
- It can handle both simple and complex relationships.
- It combines multiple decision trees.
- It generally provides good classification performance.
- It is less prone to overfitting than a single decision tree.
- It can provide feature importance information.

4.5 Train the Model

```
model.fit(X_train, y_train)  
  
print("Model training completed successfully!")
```

```
Model training completed successfully!
```

4.6 Make Predictions

```
y_pred = model.predict(X_test)  
  
print("Predictions generated successfully!")
```

Output

```
Predictions generated successfully!
```

4.7 Display Actual vs Predicted

```
results = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})

results.head(15)
```

Output

Actual	Predicted
1	1
0	0
1	1
1	0
0	0
1	1
...	...

4.8 Feature Importance

```
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

plt.figure(figsize=(10,6))

importance.head(10).plot(
    kind="bar",
    color="teal"
)
```

```
plt.title("Top 10 Important Features")  
plt.xlabel("Features")  
plt.ylabel("Importance")  
  
plt.xticks(rotation=45)
```

Analysis

The feature-importance graph shows which input variables contributed most strongly to the Random Forest model's predictions. This helps us understand which factors may be more useful for predicting student performance.
